7 STRATEGY & OUTER ENVIRONMENT

ESSENCE – CHAPTERS 10 AND 11

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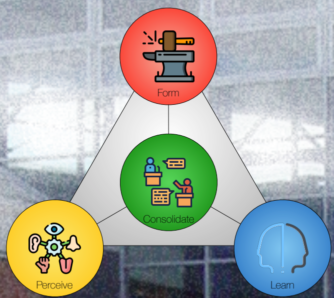
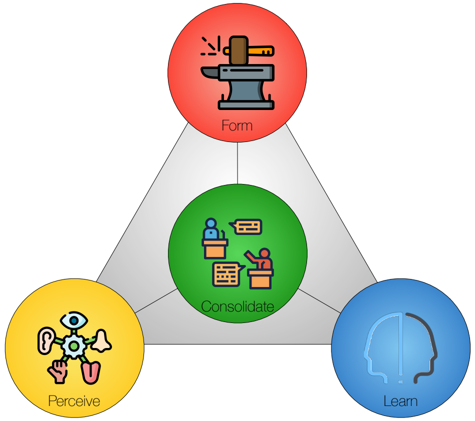


Agenda

- Exercise 6 Follow-up.
- Chapter 10: Strategy.
  - VUCA.
  - Strategic Challenges.
- Chapter 11: Outer Environment.
  - A Deweyan View on Outer Environment.
  - Four Outer Environment Categories.
  - VUCA—Outer Environment.
- Next time:
  - Exercise 7: Strategy & Outer Environment.
  - Lecture 8: Inner Environment.

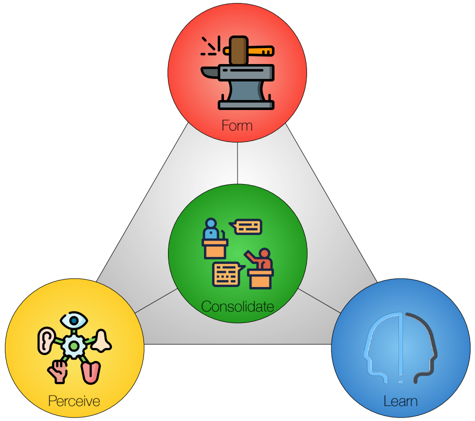


Follow-up: Exercise 6

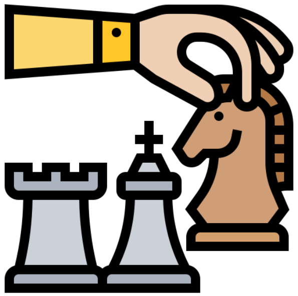
Exercise 6 (Horizon)

Use the results from Exercise 5 in this exercise. This exercise is about the project's horizon.

- Will your horizon be affected by the *technology push*? What and how?
- Will your horizon be affected by *market pull*? What and how?
- Will changes in *innovation type* characterize your horizon? How and why?
- Formulate *horizon hypotheses* as outlined in Table 9.1. Make sure you consider the future as well as the present.
- Outline some *criteria* you might use to verify or falsify your hypotheses.



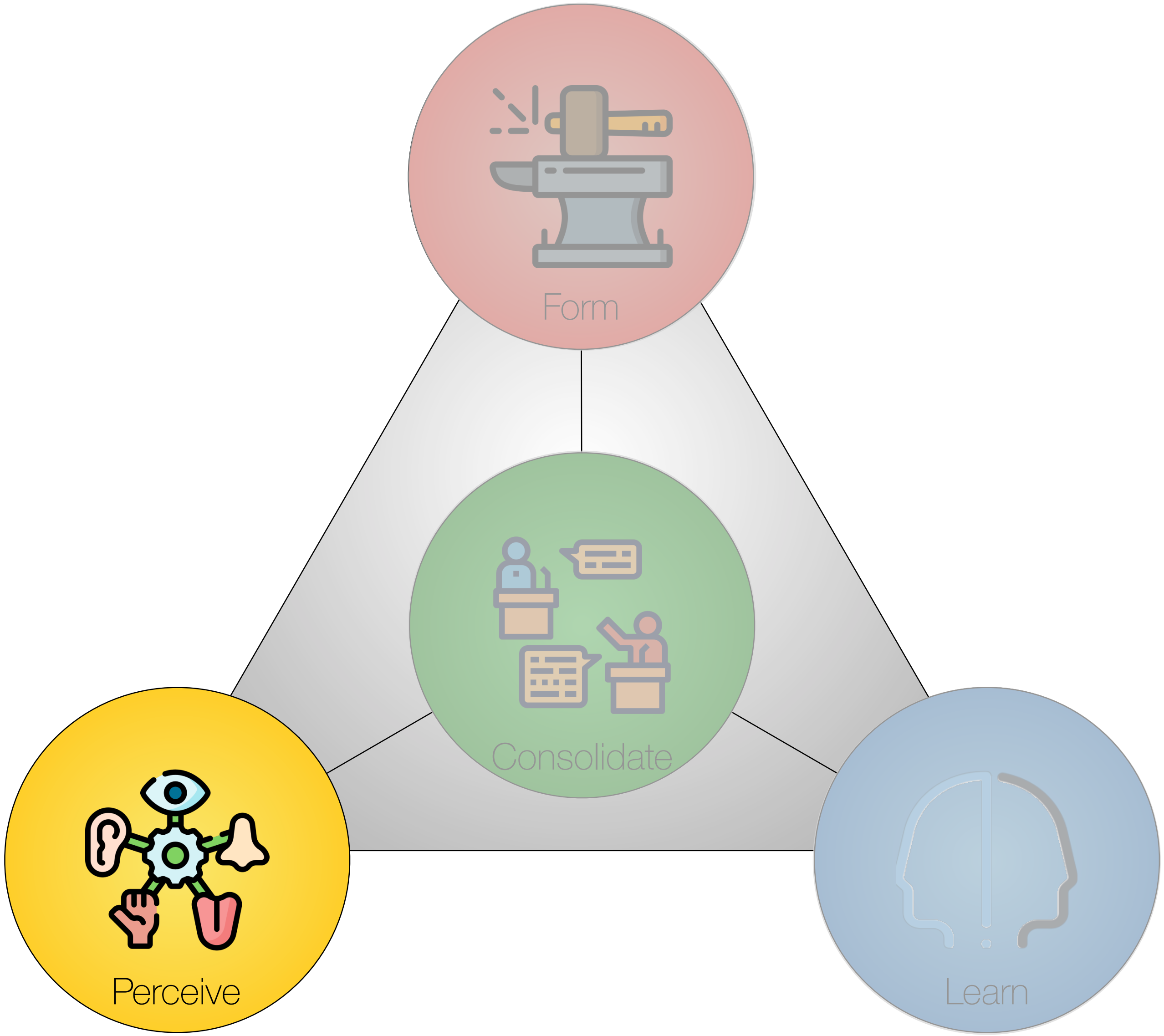
Strategy



In preparing for battle, I have always found that plans are useless, but planning is indispensable.
Dwight D. Eisenhower (1890–1969)

Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



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Three Levels of Abstraction

Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

Strategy – the overall plan for addressing the problem, including its scope and context, the essential parts needed to build the solution, and any limitations or constraints that might influence its feasibility. At the strategy level, we also evaluate the potential for future growth.

Tactics – actions planned in accordance with the strategy to achieve specific ends.

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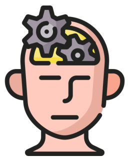
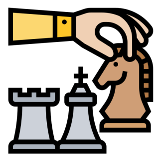
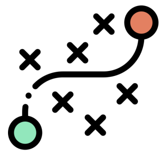
Perception Terminology







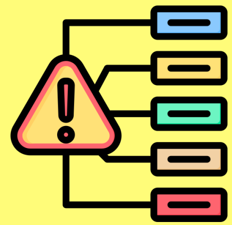
Problem

We see a problem as a situation to be handled, where problems appear as manifestations.

Outer Environment

The problem is found in the outer environment. The design is adapted to this domain.

Manifestations



Manifestations are objects, actions, or events that reflect or embody the problem.

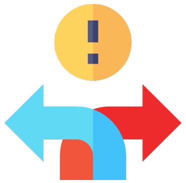
Changing Conditions

Essence is designed for conditions in which the inner or outer environment will change significantly as we learn more or as external factors affect our project.

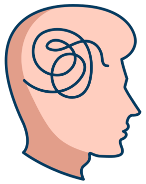
- Learning more could provide new insights about our customers, users, or ongoing projects that rely on us or on which we rely in our project.
- We cannot assume that the situation will remain constant throughout the project.
- Therefore, we cannot rely on a master plan that will remain viable throughout the project.

VUCA

- *Volatility* – i.e., conditions that change rapidly or unpredictably.



- *Uncertainty* – i.e., needs and solutions that are unsettled.



- *Complexity* – i.e., external or internal parts that form intricate meshes.

- *Ambiguity* – i.e., causes and effects that are mixed up.

Strategic Challenges for Design

		Knowledge about the situation	
Ability to predict the results of actions	High	**Complexity** External or internal parts that form intricate meshes. The situation or contribution has many interconnected parts and variables. Some information may be available, but the volume can be overwhelming.	**Volatility** Conditions that change rapidly or unpredictably. The situation or contribution is unstable or may change unexpectedly, but is not necessarily hard to understand. Knowledge about the change is often available.
	Low	**Ambiguity** Causes and effects that are mixed up. Causal relationships are completely unclear. The logic of the situation changes with our actions, and stakeholders might be unsure of their preferences.	**Uncertainty** Needs and solutions that are unsettled. Basic causes and effects are known, but there is a lack of knowledge about possible changes. The future is, therefore, unpredictable despite available information.
		Low	High

Inspired by (Bennett and Lemoine, 2014)

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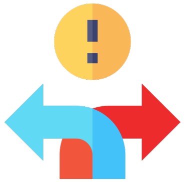
Volatility

Volatility refers to conditions that change rapidly or unpredictably.

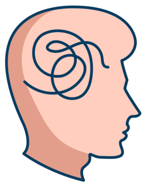
- *Outer environment:* Sudden disruptive changes in legislation, technological breakthroughs, market conditions, revenues, the economy, and competition, among other things.
- *Inner environment:* Sudden disruptive changes in leverage points, such as the technologies, components, information, or people available for building contributions.



Uncertainty

Uncertainty refers to unsettled needs and solutions.

- *Outer environment:* Unsettled disruptive changes in legislation, technological breakthroughs, market conditions, revenues, the economy, and competition, among other things.
- *Inner environment:* Unsettled disruptive changes in leverage points, such as the technologies, components, information, or people available for building contributions.



Complexity

Complexity occurs when external or internal parts of a system form intricate meshes. Complex systems are made up of a large number of parts that interact in a nonsimple way (Simon, 1962, p. 468).

- *Outer environment:* The situation involves many elements (objects or events) that interact in non-simple ways and where the properties of the whole can not easily be inferred.

- *Inner environment:* The contribution involves many elements (objects or events) that interact in non-simple ways and where the properties of the whole can not easily be inferred.

Ambiguity

Ambiguity describes challenges in which causes and effects are mixed up. Decisions or predictions are difficult to make when actions change the logic of the system in which they occur.

- *Outer environment:* The logic, needs, and constraints of the outer environment are unknown and unavailable, or causal relationships are completely unclear.
- *Inner environment:* The contribution changes the logic, needs, and constraints of the outer environment or leads to unforeseeable changes through development and implementation.

Complex Systems

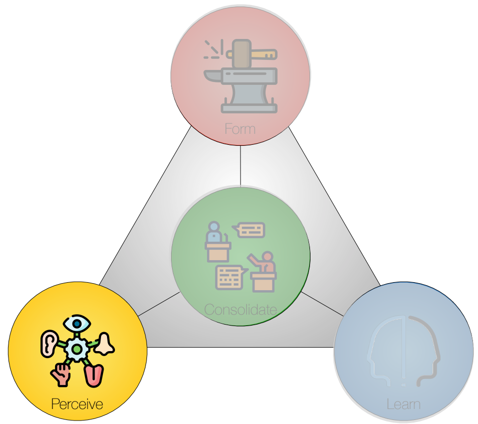
Simon says

- A complex system *is made up of a large number of parts that interact in a nonsimple way*, and it is not a trivial matter to deduce the properties of the whole from the properties of the parts.

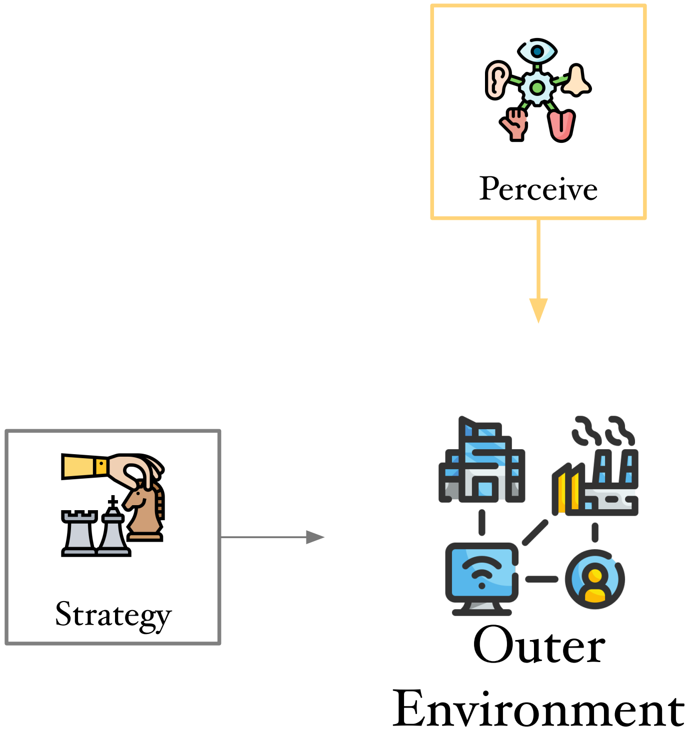
- Many complex systems have *a nearly decomposable, hierarchic structure, and this quality enables us to understand, to describe, and even to "see" such systems and their parts.*

- Such *systems will evolve from simple systems much more rapidly if there are stable intermediate forms than if there are not.*

Simon, H. A. (1962). The Architecture of Complexity. *Proceedings of the American Philosophical Society*, *106*(6), 467-482.



Outer Environment



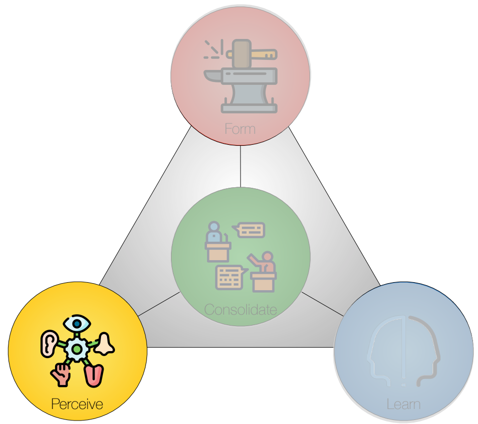
We are happy when for everything inside us there is a corresponding something outside us.

William Butler Yeats (1865 - 1939)

The Outer Environment

Our environments will not be *given* but *made*, and increasingly, they will be made independently of us by somebody else in the shape of new technologies, systems, services, sources of information, global or national standards, etc.

- In many cases, our environments will be formed or constructed by the actions of others.
- In this sense, our environment is unknown from the outset, yet we will need to start the project based on assumptions about it.



A Deweyan View on Outer Environment

The Constituents of a Problem

- The constituents are the components of a problem that allow the inquiry to proceed.
- Settling on a situation's constituents is at the core of determining the relationship between the problem and the solution.
- Constituents are conceived or selected through our appreciative system – i.e., by our insights and inclinations.
- We may, therefore, overlook constituents because we have no idea of their very existence or relevance.

Elements

In our attempts to make sense of a situation, we form concepts:

- What elements (objects or events) do we see?
- How do we describe their qualities?
- Which categories do those elements belong to?
- Elements are what we see as belonging to the situation.
- The whole is the sum of elements we see as part of the situation – the scope deemed relevant by us.

Existential and Ideational

Some elements are *existential*—they exist in the real world—and some are *ideational*.

- For example, the aisles and exits in an assembly hall are existential.
- In contrast, if a fire breaks out, our notion of an escape route is ideational and depends on where we believe the fire to be located and how we think the audience might move to escape it.
- Existential and ideational elements form part of our subsequent inquiries and may change because of them.

Old objects may undergo modification through the tests which are put upon them in new problems

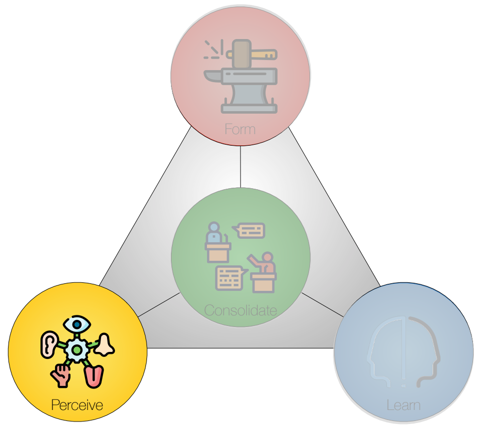
(Dewey, 1938, pp. 520-521).

Objects and Events

- Objects and events are constituents of a situation, and this situation is not a simple grouping of objects and events but a context that is experienced as meaningful.
- An object denotes a thing with properties and effects.
- In addition, objects are concepts, part of our language, that reflect what we have learned through experience.
- Dewey uses the word events to describe transitory objects that can be represented—like any object—by nouns or substantives.
- To Dewey, events have substance. Events are characterized by *a delimiting beginning, an interval, and a termination.*

Four Outer Environment Categories

Categories	Characteristics
External services	Services that are used without modification and reduce what is left for us to deal with.
External implements	Equipment, sensors, or actuators available in the outer environment and accessed via trivial interfaces.
External repositories	Repositories for collecting, storing, disseminating, and/or aggregating information via trivial interfaces.
External people	Generally available people, professionals, stakeholders, authorities, networks, organizations, etc.



VUCA

Outer Environment

VUCA Strategies for Outer Environment

Ability to predict the results of actions		Knowledge about the situation	
	High	**Complexity** External or internal parts that form intricate meshes. Strategies • Fewer manifestations • Fewer user categories • Fewer dependencies	**Volatility** Conditions that change rapidly or unpredictably. Strategies • Stockpile resources. Include force majeure clauses • Act, sense, and respond
	Low	**Ambiguity** Causes and effects that are mixed up. Strategies • Probe, sense, and respond • Look for patterns to suggest trends and sense-making	**Uncertainty** Needs and solutions that are unsettled. Strategies • Prepare contingency plans • Isolate dependencies
		Low	High

Volatility

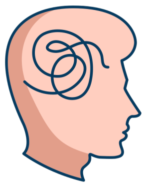
If sudden disruptive changes occur in legislation, technology, markets, revenues, the economy, or competition, among other things, we might prepare for the consequences and develop strategies for our response.

- Build *slack and stockpile resources* such as inventory or talent. Include *force majeure* clauses in the contract.
- *Act, sense, and respond* (Snowden and Boone, 2007, p. 74):
    - Act to establish order.
    - Sense where stability is present and from where it is absent.
    - Respond by transforming the situation from chaos to complexity (where identifying emerging patterns can help prevent future crises and discern new opportunities).

Uncertainty

If we face *disruptive changes* in legislation, technological breakthroughs, market conditions, revenues, the economy, competition, and more, where it is still unknown when or what they will entail:

- Prepare *contingency plans* for the alternatives we can see and
- Try to *isolate dependencies* so that the unsettled changes affect as few parts of a project or design as possible.



Complexity

The Outer Environment might be complex from the start of a project, or it may arise as the project progresses and the environment becomes better understood.

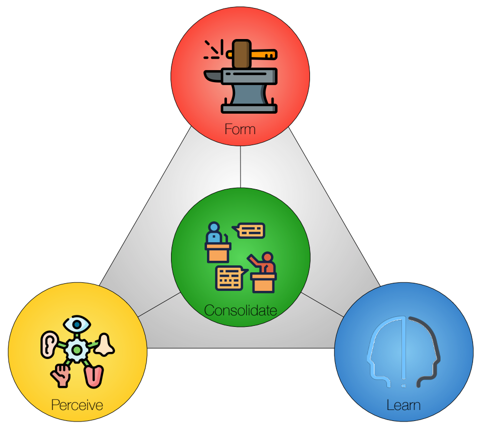
In some situations, complexity can be reduced by limitations such as:

- *Fewer manifestations.*
- *Fewer user categories.*
- *Fewer dependencies.*

Ambiguity

In the outer environment, ambiguity means that the logic, needs, and constraints are unknown and unavailable, or causal relationships are completely unclear.

- *Probe, sense, and respond* (Snowden and Boone, 2007, p. 74).
    - *Probe:* Why things happen can only be understood in retrospect.
    - *Sense:* Instructive patterns can emerge via experiments that are safe to fail.
    - *Respond:* Instead of attempting to impose a course of action, leaders must allow the path to reveal itself.
- Identify *patterns*, suggest *trends*, and *make sense* of what could be system requirements.



Next Time

Next Exercise and Lecture

- Exercise 7:
  - Strategy and Outer Environment.
- Lecture 8:
  - Chapter 12 in Essence: Inner Environment.

Exercise 7 (Strategy & Outer Environment)

This exercise is about VUCA strategies for the Outer Environment.

- Discuss VUCA challenges for your project (see Table 10.1). Which of these challenges are most likely to affect your project?
- Discuss VUCA strategies for your Outer Environment (see Table 11.2). Are they useful? How? Suggest alternative strategies if they are not useful.
- How could you use these strategies actively in your project?